

Research Statement

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Intended Advisor: Professor Youngseo Kim, UCLA Civil and Environmental Engineering

Research Vision

My long-term goal is to build an open-source, behaviorally realistic multi-agent simulation platform for transportation and urban systems. It will serve as a next-generation tool similar to MATSim or SUMO but enhanced by large language model (LLM) based agents and geospatial intelligence.

Current transportation models lack effective mechanisms for representing human emotions, intentions, and adaptability in complex networks. Although large language models have limited spatial awareness, they excel in semantic reasoning and emotional understanding. By integrating LLM-driven agents, transportation network simulations can capture individualized decision-making, context-sensitive reasoning, and passenger heterogeneity. This approach will create a more humanized and adaptable digital twin of urban mobility systems.

My proposed multi-agent simulation will enable researchers and practitioners to define population groups, demographic structures, and behavioral rules for different network environments. My goal is to integrate network accessibility and learning-augmented optimization with LLM-based agents to bridge the gap between idealized network models and real human behavior. LLM agents introduce realistic decision processes and behavioral diversity, while optimization and learning ensure measurable performance, fairness, and efficiency. Airports will serve as the primary testbed because they combine high-stakes operations, time constraints, and multi-actor coordination, but the methods can extend to urban logistics, multi-modal transportation, and emergency response in large events such as concerts, shopping festivals, or natural disasters.

Motivation and Fit

Transportation systems are complex, interconnected networks influenced by human behavior, information flow, and infrastructure design. Existing optimization and simulation methods are powerful but limited in representing the diversity and adaptability of human decision-making. Professor Kim's "HUMAN Lab," focusing on human-centered optimization, behavioral modeling, and learning reinforcement algorithms, is an ideal platform for realizing this vision. I was particularly impressed by her approach of combining econometric choice models with network optimization, which provides a theoretical foundation for linking the realism of human behavior with computational modeling.

Background

I completed my M.S. in Civil and Environmental Engineering at UC Berkeley under Dr. Jasenka Rakas, focusing on transportation network sustainability and airport systems. I use Python for large-scale spatial analysis and network modeling, and I have relied on open-source tools throughout my work. Currently, I am contributing to a CAMEL-AI project that simulates more than one million passenger agents to study adaptive route choice under disruptions in public transportation networks. This project is also testing the resilience of the public transportation network of Chicago railway while a network link is breaking. I designed the agent reasoning architecture, behavioral weighting, and network analysis modules. This project showed how integrating AI reasoning with transportation systems can replicate the diversity of real human decision processes while improving the understanding of network resilience.

Proposed Research Direction

Behavior-Aware Multi-Agent Network Modeling

Represent each traveler or vehicle as an autonomous agent within a dynamic graph constructed from GIS-based spatial data. Each agent will include behavioral attributes such as time constraints, familiarity with the environment, and adaptive reasoning ability. The framework will capture how micro-level decision-making aggregates into macro-level outcomes such as congestion propagation, delay accumulation, or uneven accessibility across a region.

Learning-Augmented Network Optimization Starting from Shortest Path Approximation

Begin with a classical shortest-path optimization framework as the benchmark for efficiency, then incorporate adaptive, stochastic, and learning-based deviations to represent realistic human decision-making. LLM-based reasoning will be used to model preference shifts under time pressure or incomplete information, particularly under disruptions or emergencies. The approach will combine reinforcement learning, convex optimization, and network flow theory to design algorithms that balance behavioral realism with computational scalability.

GIS-Integrated Multi-Agent Simulation Platform Development

Build an open-source simulation environment that integrates network analysis, behavioral modeling, and spatial datasets. Users will be able to input real-world network data through GIS, define population structures, and test customized policy or infrastructure scenarios. The platform will include modules for route planning, multimodal transfer, evacuation design, and emergency response simulation.

Benchmarking and Validation Against Established Models

Evaluate the proposed framework against existing tools such as MATSim and SUMO to ensure transparency, scalability, and validity. Performance metrics will include computational efficiency, behavioral accuracy, and adaptability in high-density or emergency situations. Benchmarking will demonstrate that the LLM multi-agent system provides a measurable contribution to the field of transportation modeling and future industry applications.

PhD Development Plan

In the first one to two years I plan to build depth in learning-augmented optimization and behavior modeling through coursework and targeted projects, while contributing to a well-defined problem suitable for an early publication. By the start of year three I expect to formulate an independent, behavior-aware network optimization agenda that ties together my airport applications with broader urban mobility questions, with the goal of producing two to three high-quality papers in top transportation and AI venues.

Closing

I look forward to the possibility of contributing to the HUMAN Lab. I am excited by the chance to develop human-centered and provably effective mobility systems with Professor Kim's guidance, and to collaborate across UCLA on projects that advance accessibility, resilience, and sustainability at scale.